

Ising Optimisation for Combinatorial Auction Clearing

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Abstract

Combinatorial auction clearing, which allocates indivisible or volume-constrained goods to bidders under multiple simultaneous supply constraints, is NP-hard in general and provably hard to approximate. We formulate the clearing problem as an Ising ground-state search/QUBO, derive a spin-flip dominance condition for correctly calibrating per-constraint penalty coefficients under heavy-tailed bid-size distributions, and use a solver-agnostic greedy repair step. We then instantiate this general formulation on the EU Emissions Trading System (EU ETS), reconstructing a synthetic bid population from published EEX auction aggregates and benchmarking simulated annealing (dwave-neal) against an exact integer programming (IP) baseline. Across 20 sampled EEX sessions, simulated annealing achieves a mean welfare gap of 1.37% against the IP optimum, remaining stable (0.64–1.13%) as the bid population is scaled up to $6\times$ its base size.

1 Problem Statement

We consider a combinatorial auction with N bids competing for volume drawn from M distinct, independently supply-constrained resource types. Each bid i requests a quantity q_i of a single type $\tau(i)$ at value v_i , and is either fully accepted or fully rejected, there is no partial fulfillment. Some pairs of bids carry a *complementarity*: joint acceptance of both is worth strictly more than the sum of their individual values, because the bidder derives value from holding a bundle across types.

Formally, let $x_i \in \{0, 1\}$ indicate acceptance of bid i . The clearing problem is

$$\max_x \sum_i v_i x_i + \sum_{(i,k) \in \mathcal{P}} w_{ik} x_i x_k \quad (1)$$

$$\text{s.t.} \quad \sum_{\{i: \tau(i)=\tau\}} q_i x_i \leq S_\tau \quad \forall \tau, \quad (2)$$

where S_τ is the supply of type τ , $\mathcal{P}$ is the set of complementary bid pairs, and $w_{ik} > 0$ is the synergy value of accepting i and k together. In the special case where each bid also requests a specific indivisible lot j (rather than fungible volume of a type), an additional exclusivity

constraint $\sum_i a_{ij} x_i \leq 1$ per lot applies, with $a_{ij} = 1$ if bid i touches lot j .

This is a weighted generalisation of k -set packing, where k is the maximum number of resource types or lots any single bid touches. Hazan, Safra and Schwartz [1] proved that no polynomial-time algorithm approximates k -set packing within $o(k/\log k)$ unless $P = NP$. This motivates a probabilistic / physics-inspired computing approach: encoding the clearing problem as an Ising ground-state search and solving it via simulated or physical annealing, rather than pursuing exact combinatorial search at scale.

2 Ising / QUBO Formulation

Each acceptance variable x_i maps to a spin $\sigma_i \in \{-1, +1\}$ via $x_i = (1 + \sigma_i)/2$, with $\sigma_i = +1$ denoting acceptance. We construct the Hamiltonian $H = H_{\text{obj}} + H_{\text{pkg}} + H_{\text{cap}}$ term by term, plus an optional H_{excl} for the indivisible-lot case.

Welfare objective.

$$H_{\text{obj}} = -A \sum_i v_i x_i. \quad (3)$$

Spins with large v_i experience a strong external field biasing them toward $\sigma_i = +1$.

Package synergy.

$$H_{\text{pkg}} = -A \sum_{(i,k) \in \mathcal{P}} w_{ik} x_i x_k. \quad (4)$$

This introduces attractive off-diagonal couplings $J_{ik} = Aw_{ik}$, making bid-acceptance decisions non-separable: the optimal choice for bid i depends on whether its complementary partner k is accepted.

Per-type supply cap.

$$H_{\text{cap}} = \sum_\tau C_\tau \left(\sum_{\{i: \tau(i)=\tau\}} q_i x_i - S_\tau \right)^2. \quad (5)$$

This penalises over-allocation of each type τ against its own supply S_τ independently, so a penalty on one type does not interact with another type's cap.

Lot exclusivity (indivisible-lot case only).

$$H_{\text{excl}} = B \sum_j \sum_{i < k} a_{ij} a_{kj} x_i x_k. \quad (6)$$

Any two bids sharing an indivisible lot j incur an antiferromagnetic coupling, penalising simultaneous acceptance. This term is only needed when the underlying resource is a discrete, uniquely-owned lot; for a fungible, volume-constrained resource it is redundant with H_{cap} below and should be omitted (Section 5).

Substituting $x_i = (1 + \sigma_i)/2$ throughout yields the standard Ising form

$$H = - \sum_i h_i \sigma_i - \sum_{i < k} J_{ik} \sigma_i \sigma_k + \text{const}, \quad (7)$$

whose ground state corresponds to the optimal penalised allocation.

3 Penalty Coefficient Derivation

3.1 Why a single global coefficient fails

The squared penalty $C(\sum_i q_i x_i - S_\tau)^2$ is structurally problematic when $q_i \ll S_\tau$. Whenever many small bids compete for shared supply. Expanding the square gives a diagonal term $C(q_i^2 - 2S_\tau q_i)$, which is net negative for all $q_i < 2S_\tau$. The supply constraint is therefore enforced only through accumulated *off-diagonal* terms as more bids of the same type are accepted, requiring a many-body dominance bound rather than the standard single-violation condition $C > A v_{\text{max}}$.

3.2 Spin-flip dominance bound

Consider a solver evaluating whether to accept bid i given n_{acc} bids of the same type already accepted. The energy change on flipping x_i is

$$\Delta E = Q_{ii} + 2 \sum_{k \in \text{acc}} Q_{ik}. \quad (8)$$

For the flip to be correctly rejected when it would violate the constraint ($\Delta E > 0$), accumulated off-diagonal pushback from H_{cap} must exceed the combined reward from H_{obj} and any package coupling:

$$2Cq_i \sum_{k \in \text{acc}} q_k > Av_i + Aw_{ik} \mathbf{1}_{\{k \in \mathcal{P}_i\}} \quad (9)$$

Taking the worst case $v_i = v_{\text{max}}$, $w_{ik} = w_{\text{max}}$, $\sum_k q_k \approx n_{\text{acc}} \bar{q}$, and normalizing by S_τ so $S_\tau \rightarrow 1$:

$$C > \frac{A(v_{\text{max}} + w_{\text{max}})}{2q_i n_{\text{acc}} \bar{q}}. \quad (10)$$

3.3 Per-type coefficients

A single coefficient derived from a pooled mean $\bar{q}$ across all types under-penalises any type whose bid-size distribution is particularly heavy-tailed, e.g. a type containing one bid that consumes a large fraction of its own supply on its own. We therefore derive a coefficient C_τ per type, applying Eq. (10) with each type's own v_{max}^τ , w_{max}^τ , and q_{max}^τ as the worst-case denominator – binding on the single largest bid in each type, which is the tightest constraint under a heavy-tailed distribution:

$$C_\tau = C_{\text{mult}} \cdot \frac{v_{\text{max}}^\tau + w_{\text{max}}^\tau + 1}{2q_{\text{max}}^\tau}. \quad (11)$$

The QUBO matrix applies each type's C_τ only to its own capacity-penalty block; H_{obj} and H_{pkg} are unaffected.

4 Feasibility Enforcement

Penalty-based QUBO encoding enforces constraints only softly. Even with a well-calibrated C_τ , heavy-tailed bid distributions occasionally produce feasibility violations at the energy-landscape boundary, where accumulated off-diagonal pushback marginally fails to outweigh the objective reward for the largest bids.

A static supply buffer ($S_\tau \rightarrow (1 - \epsilon)S_\tau$ in the QUBO) can absorb this, but is solver-specific: any such margin is tuned against one solver's particular annealing dynamics and has no principled basis for a different solver or sweep schedule.

We instead apply a solver-agnostic classical post-processing repair. After the QUBO solver returns a candidate solution x , feasibility is checked per type. Where violated, a greedy knapsack repair is applied: accepted bids in the infeasible type are ranked in descending order of value density v_i/q_i , and bids are removed from the bottom of this ranking until the constraint is satisfied:

$$\text{drop} = \arg \min_{i \in A_\tau} \frac{v_i}{q_i}, \quad \text{until} \quad \sum_{i \in A_\tau} q_i \leq S_\tau. \quad (12)$$

This operator is independent of both the QUBO construction and solver choice. Its welfare cost is bounded by the value density of the dropped bids, which are by construction the least valuable per unit of supply consumed. The true supply cap S_τ (no buffer) is used in all feasibility checks reported below.

5 Application: EU ETS Carbon Credit Clearing

We instantiate the formulation above on the EU Emissions Trading System (EU ETS), under which the European Commission distributes a fixed system-wide cap of allowances across general allowances (EUA), aviation allowances (EUAA), and member-state sub-quotas for 25

participating states. From 2025, EUAA volumes are integrated into EUA spot auctions, creating explicit cross-type allocation decisions. All economically relevant inputs demand values, per-type supply, and type assignments are fixed by published regulatory decisions [5]; integer programming serves as the natural exact classical baseline throughout.

5.1 Data and bid reconstruction

Session statistics are taken from the EEX Primary Auction Spot Report 2025 (Here), covering 213 sessions across four credit types (EUA, EUA_DE, EUA_PL, EUAA) over 2025, with per-session clearing price, supply volume, bid count, mean and standard deviation of bid volume, and cover ratio. Individual bid-level data is confidential under EU ETS regulations, so we reconstruct a synthetic bid population whose statistics match the published aggregates exactly, standard practice in computational economics when microdata is unavailable [6]. Bid quantity q_i is drawn log-normal, with parameters fit to the published mean μ_q and standard deviation σ_q via moment matching:

$$\mu_{\ln} = \ln \left(\frac{\mu_q^2}{\sqrt{\sigma_q^2 + \mu_q^2}} \right), \quad \sigma_{\ln} = \sqrt{\ln \left(1 + \frac{\sigma_q^2}{\mu_q^2} \right)}. \quad (13)$$

Log-normal is a standard and tractable choice for bid-size data: it is supported on $(0, \infty)$, guarantees positivity without truncation, and its two parameters are uniquely determined by (μ_q, σ_q) for any observed coefficient of variation, unlike heavier-tailed alternatives (e.g. Pareto) whose method-of-moments fit is only well-posed when $\sigma_q/\mu_q > 1$ and becomes numerically unstable as this ratio grows. Bid price is triangular, $p_i \sim \text{Triangular}(p_{\min}, p^*, p_{\max})$, with mode at the published clearing price p^* ; $v_i = q_i p_i$. Quantities are rescaled so total bid volume matches the real session cover ratio $\rho = \sum_i q_i / S_\tau$ (observed range 1.54–2.06). A single representative (median) session per type is used, yielding 321 reconstructed bids: EUA 100, EUA_DE 90, EUA_PL 82, EUAA 49.

5.2 Mapping onto the general formulation

H_{excl} is dropped. EU ETS allowances are fungible tCO₂ volume drawn from a common pool, not discrete lots exclusive to one bidder. Any number of bids may be accepted provided total accepted volume does not exceed S_τ ; this is enforced entirely by H_{cap} . Retaining a lot-exclusivity term (as would be needed in the general indivisible-lot case) here imposes thousands of spurious antiferromagnetic couplings across same-type bids and prevents the solver from accepting more than a handful of bids regardless of true supply headroom.

H_{pkg} models cross-type compliance bundles. Aviation operators holding EUAA allowances often also participate in EUA sessions to cover non-aviation installations; such a bidder values an EUA–EUAA bundle jointly more than the sum of its parts, since holding only one type leaves a compliance gap. We set $w_{ik} = 0.15 \min(v_i, v_k)$ and model 30% of EUAA bidders as holding a linked EUA bid, giving 14 complementary pairs in the toy instance.

Per-type C_τ . The four EU ETS types exhibit markedly different bid-size heavy-tailedness:

Type	n	$\bar{q}$	$q_{\max}$	$\sum q$
EUA	100	0.0139	0.2859	1.39
EUA_DE	90	0.0242	0.6524	2.18
EUA_PL	82	0.0210	0.2296	1.72
EUAA	49	0.0337	0.3258	1.65

Table 1: Per-type bid-quantity statistics (normalised) for example case.

EUA_DE contains a single bid consuming 65.2% of that type’s supply a large industrial operator’s annual block bid for which a pooled-mean coefficient would be $\sim 30\times$ too weak. With $C_{\text{mult}} = 100$, the resulting C_τ are:

Type	C_τ
EUA	352.5
EUA_DE	154.5
EUA_PL	438.9
EUAA	309.3

Table 2: Per-type penalty coefficients.

5.3 Solvers

Two solvers are benchmarked on the same QUBO instance. **Integer Programming (IP):** PuLP/CBC branch-and-bound, solving the problem exactly with native hard supply constraints (package terms linearised via auxiliary variables z_{ik} with McCormick constraints); this is the welfare optimality ceiling. **Simulated Annealing (SA):** `dwave-neal` [7], 200 independent reads, 1,000 sweeps per read, on the per-type- C_τ QUBO with greedy repair post-processing. All quantities are normalised per type before building the QUBO; welfare is reported in real euros from raw bid values.

5.4 Results

Multi-session benchmark. Across 20 randomly sampled EUA sessions (each paired with fixed median EUA_DE, EUA_PL, EUAA sessions), SA returns a feasible solution on every session after repair.

SA achieves a mean welfare gap of 1.11% against the exact IP optimum across the 20 sessions, with the gap rang-

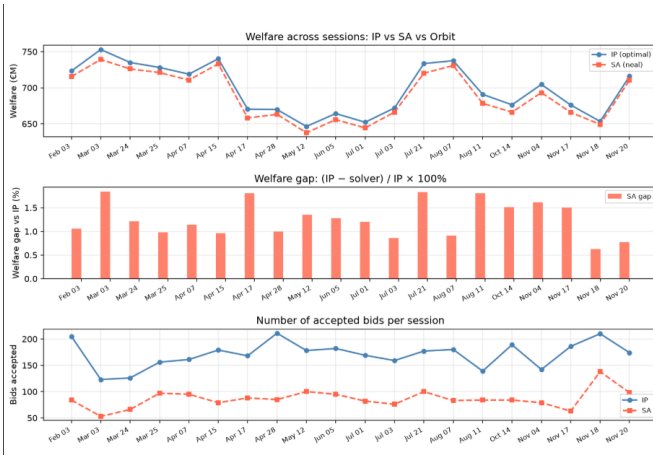


Figure 1: Welfare gap (%) vs. IP across 20 sampled EUA sessions, with per-session number of accepted bids and feasibility status.

Solver	Mean gap (%)	Range (%)
SA (neal)	1.11	0.63 – 1.84
IP	0.00	—

Table 3: Welfare gap vs. IP across 20 sessions (mean and range).

ing from 0.63% to 1.84% (Table 3). This tight, consistently low range rather than a small mean dragged down by a few near-perfect sessions against otherwise poor ones indicates that the per-type penalty coefficients C_τ and greedy repair generalise across sessions with substantially different clearing prices, cover ratios, and bid-size distributions, rather than being overfit to a single calibration session. Every session returns a fully feasible allocation after repair (Section 4), so the reported gap reflects only the welfare cost of the annealing search itself and the small amount of value lost to repair, not any residual constraint violation. Figure 1 shows this gap alongside the number of bids accepted per session by each solver, confirming that SA does not systematically under-accept relative to IP the welfare shortfall comes from marginally suboptimal bid selection near the supply boundary, not from excessive conservatism. vs. IP across 20 sessions.

Scaling benchmark. Runtime and optimality gap across problem sizes $n \in \{321, 481, 642, 802, 963, 1123, 1284, 1444, 1605\}$ (base bid population scaled 1–6 $\times$):

IP runtime grows super-linearly, consistent with the exponential worst-case complexity of branch-and-bound. SA grows sub-linearly. SA’s optimality gap stays stable at 0.64–1.13% across all sizes, indicating the per-type- C_τ formulation with greedy repair scales without quality degradation.

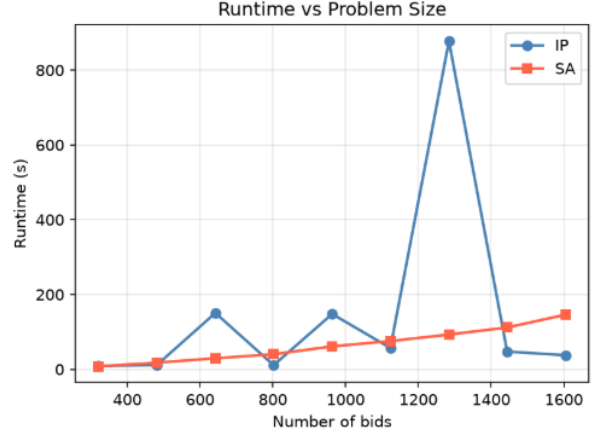


Figure 2: BenchMark Results for Scale Testing

6 Discussion and Extensions

6.1 Constrained Quadratic Model as a scaling path

D-Wave’s Constrained Quadratic Model (CQM) separates the objective from the constraints, stating $\sum q_i x_i \leq S_\tau$ as a native inequality and enforcing it exactly without the diagonal imbalance of Section 3. A slack-variable encoding was investigated as an intermediate approach but fails for the same structural reason: slack bits must span $[0, 1]$ in normalised units, so the largest bit (≈ 0.5) is $\sim 35\times$ larger than a typical bid coefficient, reproducing the scaling imbalance elsewhere; uniform slack spacing avoids this but requires $\sim 10,000$ bits per type, making the QUBO intractable. CQM (via `LeapHybridCQMSampler`) is identified as the natural path to scale beyond the toy instance.

6.2 Extensions within the current framework

(1) A 9-type model treating each EEX fund pool (IF, IX, MF, MX, SF) as a separate type τ ; (2) degree-3 package terms for three-way EUA–EUAA–EUA.DE bundles via Boros–Hammer [4]; (3) multi-session calibration across all 213 sessions to test parameter stability over the observed price range of €70–84 per tCO₂.

Code Availability

https://github.com/satkarjuneja/Carbon_Credit_Auction_Clearing

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